EE3032 Introduction to VLSI Design—Homework 4

1. Fig. 1 shows the schematic diagram of a two-input CMOS AND gate driving a 1fF capacitance. Determine the transistor sizes arbitrarily and complete the following questions.

   (a) Simulate your design using Hspice simulator and measure the worst-case high-to-low and low-to-high propagation delays at node Z and F. (homework 3)

   (b) Draw the layout the the AND gate and extract the netlist. Then, measure the worst-case high-to-low and low-to-high propagation delays at node Z and F. (10 points)

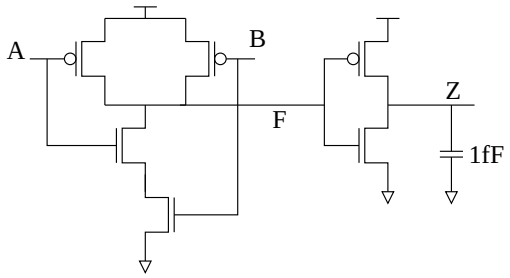

Figure 1: An AND gate.